

9_pop quiz demos, Coulombs

 This is a preview of the published version of the quiz

Started: Mar 22 at 7:09pm

Quiz Instructions

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Question 1 20 pts

If lightning strikes your car

(hint: its a Faraday cage)



charges will pread inside the car and outisde the car



charges will move inside the car



charges will spreadout on the outside chassis of the car



charges will stay wherever the lightning stoke



Question 2 20 pts

If you have 2 charges q_1 and q_2 attracting each other with a force of 10N . You keep the distance the same but you triple $q_2=q_2'$ and keep q_1 the same $q_1'=q_1$. The new force between q_1' and q_2' is



about 1N



30N



about 3N



90N



Question 3 15 pts

A metallic object holds a charge of -3.8×10^{-6} C. What total number of electrons does this represent? ($e = 1.6 \times 10^{-19}$ C is the magnitude of the electronic charge.)

☐

2.4×10^{13}

☐

1.6×10^{14}

☐

6.1×10^{13}

☐

4.2×10^{14}



Question 4 15 pts

Two point charges are 4 cm apart. They are moved to a new separation of 2 cm. By what factor does the resulting mutual force between them change?

☐

2

☐

$1/4$

☐

$1/2$

☐

4



Question 5 15 pts

If the size of the charge value is tripled for both of two point charges maintained at a constant separation, the mutual force between them will be changed by what factor?

hint: the force is proportional to the product of the charges

☐

$1/9$

☐

0.33

☐

9

☐

3



Question 6 5 pts

You rub a balloon against your hair and now the balloon sticks to the wall. The wall is an insulator.

This is because of:

☐

saturation

☐

electrolysis

☐

polarization

☐

oxydation

☐

Question 7 5 pts

It is said that electric charge is quantized, which means that the charge on an object

☐

is a whole-number multiple of the charge of one electron.

☐

may occur in an infinite variety of quantities.

☐

can be neither created nor destroyed.

☐

will interact with neighboring electric charges.

☐

is sometimes positive.

☐

Question 8 5 pts

As discussed in class.

A Van graaf generator can be raised to large voltage. 1000 V or more. However, it can shock you but it wont kill you. That's because

(supposed you are not isolated from the ground)

☐

Its spherical in shape

☐

The charges can not be transfered to your body if you touch it

☐

Your body is an insulator.

☐

It can store only a tiny amount of charge (coulombs)

☐

Question 9 2 pts

3 nC is

☐

$3 \cdot 10^{-9} \text{C}$

☐

$3 \cdot 10^3 \text{C}$

☐

$3 \cdot 10^6 \text{C}$

☐

$3 \cdot 10^{-6} \text{C}$



Question 10 2 pts

Coulombs law says that the electric force between two charges placed at a distance r from each other is

☐

inversely proportional to the distance

☐

inversely proportional to the distance squared

☐

proportional to the distance

☐

is proportional to the distance squared

☐

does not depend on the distance

Not saved

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